

# Braid Presentations for Concurrent Programs

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## Abstract

Ziemiański’s cube chain category  $\text{Ch}(K)$  is a combinatorial model for the space of directed paths on a precubical set  $K$ . Paliga and Ziemiański built a braid invariant for precubical sets by observing a relation between  $\pi_1|\text{Ch}(Z)|$  and braids, where  $Z$  is the terminal precubical set. However, that invariant trivializes for Euclidean  $K$  because  $|\text{Ch}(I^n)|$  is contractible [PZ21, Prop. 6.1]. This paper constructs a localization of  $\text{Ch}(K)$  at morphisms that preserve the order of coordinates, called  $\text{Br}(K)$ . And we will show that this structure is the source of the braid structure of  $\pi_1|\text{Ch}(Z)|$ . But, there is no nerve or geometric realization in this technique. Instead we build a 2-polygraph giving an artin braid-like presentation of  $\text{Br}(K)$ . The  $\{0, 1, 2\}$ -cells of the polygraph are the dimension  $\{0, 1, 2\}$  chains of  $K$ . Intuitively, 3-cells of  $K$  contribute braid relations, and pairs of non-intersecting 2-cells of  $K$  contribute commuting relations. This is not just a metaphor: at  $K = Z$ , we get the artin relations and generators exactly. So  $\text{Br}(Z)$  is equivalent to the positive braid category. The presentation is natural in  $K$ , providing a generalization of the existing work. As an application, we establish a link to the hyperplane arrangement literature. The way braids are constructed via Salvetti’s theorem for the braid arrangement exactly matches the polygraph we build here for the cube. All this work is formalized in Lean [Sto26].

## 1 Introduction

Ziemiański’s cube chain category  $\text{Ch}(K)$  models the executions of a bipointed precubical set  $K$  [Zie20, Thm. 7.6], and Paliga–Ziemiański’s homotopy equivalence between  $|\text{Ch}(Z)|$  and configuration spaces of points in the plane makes it a braid invariant of any  $K$ , by naturality. It trivializes on Euclidean complexes [PZ21, Prop. 6.1] for a one-line reason: it factors through the contractible  $|\text{Ch}(I^n)|$ .

The braiding is already present one level down, in  $\text{Ch}(K)$  itself, with no nerve. An object of  $\text{Ch}(K)$  cuts an execution of  $N$  events into consecutive blocks of simultaneous ones; a morphism is a *refinement*, a relabelling of the events that coarsens the blocking. Some refinements only glue adjacent blocks and leave every event at its own position; the rest genuinely reorder events across a block boundary. Call the first kind *order preserving*, write  $W$  for the class they form, and set  $\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}]$ . Nothing is quotiented: inverting  $W$  leaves a category whose arrows remember which events crossed which, and how often. That is a braiding, not a permutation — two independent events that swap twice are not the identity.

We present  $\text{Br}(K)$  by generators and relations, for every  $K$ , naturally in  $K$ . The generators and relations are not invented; they are the low-dimensional chains of  $K$ . A chain of degree 0 is a *run*, an execution with a total order on its events, and the runs over a chain  $e$  form a poset with a least element  $\perp_e$  and a greatest element  $\top_e$ : the run  $e$  is glued from, and the run that reverses every block

of  $e$ . So a chain is already arrow-shaped, from the source of  $\perp_e$  to the source of  $\top_e$ , and we take the low-dimensional ones as cells. Write  $\mathcal{P}(K)$  for the resulting 2-polygraph (Definition 4.14): 0-cells the runs, 1- and 2-cells the chains of degree 1 and 2, a 2-cell  $e$  attached along the two maximal chains of the poset of runs over  $e$ .

**Theorem 1.1** (Theorem 5.4, Theorem 6.1, Theorem 7.6). *Let  $K$  be a bipointed precubical set.*

1.  $\mathcal{P}(K)$  presents  $\text{Br}(K)$ .
2.  $K \mapsto \mathcal{P}(K)$  is a functor  $\square_* \rightarrow \mathbf{Poly}$ , and the presentation is natural in  $K$  as an equality of functors.
3.  $\mathcal{P}(Z) \cong \mathcal{P}_{\text{Artin}}$ , cell for cell and word for word; hence  $\text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$ .

*No hypothesis on  $K$  is needed for any of the three.*

Artin’s dichotomy is the dichotomy between the two kinds of degree-two chain. One block of dimension 3 spans a hexagon of runs and gives  $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ ; two blocks of dimension 2 span a square and give  $\sigma_i \sigma_k = \sigma_k \sigma_i$ . So 3-cells of  $K$  contribute braid relations and disjoint pairs of 2-cells contribute commutations. The closest existing statement is Gaussent–Guiraud–Malbos’s coherent presentation of an Artin monoid [GGM15], obtained from Garside theory; here the same 2-cells come from the geometry of  $K$ , and the Artin case is the value at a point. Nothing about braids is used to prove Theorem 1.1(iii): the generators come out of  $\text{Ch}(Z)$  as its degree-one chains and the relations as its degree-two chains.

**Formalisation.** All of the below is formalised in Lean 4 on mathlib [Sto26], sorry-free. The objects  $\text{Br}(K)$ ,  $W$  and  $\mathcal{P}(K)$  are `ChainCat.Br`, `ChainCat.W` and `Paper.poly`, and the three theorems are `Paper.paperPresents`, `Paper.paperSquare` and `Paper.paperArtinIso`. The file `Paper.lean` is this paper read back into the tree: each statement below appears there as an `example` discharged by one citation.

**Notation.** **Cube** is the box category,  $\square = \widehat{\mathbf{Cube}}$  the precubical sets,  $\square_*$  the bipointed ones with  $Z$  final, and  $I^n$  the  $n$ -cube, bipointed by  $(0, \dots, 0)$  and  $(1, \dots, 1)$ . We set  $[n] := \{1, \dots, n\}$  and identify  $K_n = \text{Hom}_{\square}(I^n, K)$  by Yoneda. Ambient categories are bold, constructions applied to an object upright, posets and polygraphs calligraphic. Composition is classical:  $gf$  means “first  $f$ , then  $g$ ”. A word in a polygraph is a path read left to right, so  $\sigma_i \sigma_k$  is the arrow  $\sigma_k \circ \sigma_i$ ; this, together with the convention of [PZ21; Zie20] that a refinement runs from a finer chain to a coarser one, is why two  $^{\text{op}}$ s appear, in  $\text{Br}(K)$  and in  $\mathbf{Braid}^{+, \text{op}}$ . Both are bookkeeping.

## 2 Background

**Cube chains** [PZ21; Zie20]. The **wedge**  $P \vee Q$  of  $P, Q \in \square_*$  is the pushout in  $\square$  gluing  $\text{fin}_P$  to  $\text{init}_Q$ , bipointed by  $\text{init}_P$  and  $\text{fin}_Q$ ;  $(\square_*, \vee, I^0)$  is monoidal. A **shape** is a list  $a = [a_1, \dots, a_r]$  of positive integers, with **serial wedge**  $\vee a := I^{a_1} \vee \dots \vee I^{a_r}$  and **blocks** the  $I^{a_i}$ ; put  $|a| := \sum_i a_i$  and  $\deg a := \sum_i (a_i - 1) = |a| - r$ . Then  $\text{Ch}(K)$  is the full subcategory of  $\square_*/K$  spanned by the serial wedges: an object is a bipointed map  $\vee a \rightarrow K$ , and a morphism, called a **refinement**, is a map of wedges commuting over  $K$ . We write  $\deg d$  and  $|d|$  for the degree and event count of a chain’s shape, and  $\text{Ch}(u) : \text{Ch}(K) \rightarrow \text{Ch}(L)$  for the pushforward along  $u : K \rightarrow L$ , which postcomposes the classifying map and leaves the shape alone. A chain of degree 0 is a **run**; the full subcategory  $\text{Run}(K)$  of runs is discrete, since a refinement that loses no block is an identity.

We use three facts from [PZ21, §2]. Serial wedges are non-self-linked and admit an altitude function  $\text{alt} : K_0 \rightarrow \mathbb{Z}$  with  $\text{alt}(\text{fin } e) = \text{alt}(\text{init } e) + 1$  on edges, the final vertex of  $\bigvee a$  having altitude  $|a|$ . A bipointed map between connected objects admitting altitude preserves it [PZ21, Prop. 2.7]. And  $\text{Ch}$  is lax monoidal from  $(\square_*, \vee)$  to  $(\mathbf{Cat}, \times)$  by concatenation, with  $\text{Ch}(P) \times \text{Ch}(Q) \rightarrow \text{Ch}(P \vee Q)$  an isomorphism when  $P \vee Q$  is non-self-linked and admits altitude [PZ21, Lem. 2.11].

**The events of a shape.** Let  $\text{Coord} : \mathbf{Cube} \rightarrow \mathbf{Set}$  send  $I^n$  to  $[n]$  and a face embedding  $g$  to the monotone injection  $g_*$  naming the directions  $g$  leaves free, and let  $E(\text{Coord})$  be its left Kan extension along Yoneda. That functor is cocontinuous and  $\text{Coord}(I^0) = \emptyset$ , so the pushout defining a wedge becomes a coproduct:

$$\text{Ev}(a) := E(\text{Coord})\left(\bigvee a\right) \cong \coprod_i [a_i].$$

We call  $\text{Ev}(a)$  the **events** of  $a$  and write  $f_* := E(\text{Coord})(f)$ .

**Convention 2.1.** We identify  $\text{Ev}(a)$  with  $[N]$ ,  $N = |a|$ , by listing  $\coprod_i [a_i]$  block by block, and write  $\text{blk}_a : [N] \rightarrow [r]$  for the resulting block map. So a shape on  $N$  events is precisely a partition of  $[N]$  into consecutive intervals, and it enters everything below only through  $\text{blk}_a$ . We drop the subscript when the shape is clear.

**Proposition 2.2.** *Let  $f : \bigvee a \rightarrow \bigvee b$  in  $\square_*$ , with  $a$  and  $b$  shapes on  $N$  and  $M$  events. Then  $N = M$ , the map  $f_*$  is a permutation of  $[N]$  determined by and determining  $f$ , and there is a monotone  $\text{blk}(f)$  with*

$$\text{blk}_b \circ f_* = \text{blk}(f) \circ \text{blk}_a, \tag{1}$$

*the restriction of  $f_*$  to each block of  $a$  being increasing.*

*Proof.* By the pushout,  $f$  is determined by its restrictions to the blocks of  $a$ , each of which lands in a single block of  $b$  because  $a_i > 0$ ; that gives  $\text{blk}(f)$ , equation (1) and the last clause, and a face embedding is determined by the directions it leaves free, so  $f_*$  determines  $f$ . Injectivity of  $f_*$  and monotonicity of  $\text{blk}(f)$  follow by induction on the length of  $b$ : by [PZ21, Lem. 2.11] a map  $\bigvee a \rightarrow I^m \vee \bigvee b'$  splits as  $f_1 \vee f_2$  over a splitting of  $a$ , so  $f_*$  is a disjoint union and  $\text{blk}(f)$  is  $\text{blk}(f_1) \sqcup (1 + \text{blk}(f_2))$  with  $\text{blk}(f_1)$  constant at 1; injectivity of the left half is a second induction, the two images at each glued vertex being separated by the coordinates already equal to 1 there. Finally  $f$  preserves altitude [PZ21, Prop. 2.7], so  $N = M$  and the injection  $f_*$  is a permutation.  $\square$

**Presented categories.** A **2-polygraph**  $\mathcal{P}$  consists of a set  $V$  of 0-cells, families  $\mathcal{P}_1(x, y)$  and  $\mathcal{P}_2(x, y)$  of 1- and 2-cells for  $x, y \in V$ , and two maps  $\text{src}, \text{tgt} : \mathcal{P}_2(x, y) \rightarrow \{\text{paths from } x \text{ to } y\}$  into the free category on the 1-cells; we call  $\text{src}(\alpha)$  and  $\text{tgt}(\alpha)$  the two **words** of  $\alpha$ . A morphism of polygraphs is a map in each dimension commuting with  $\text{src}$  and  $\text{tgt}$  on the nose, and **Poly** is the resulting category. The category  $\langle \mathcal{P} \rangle$  **presented by**  $\mathcal{P}$  is the free category on the 1-cells modulo the congruence generated by  $\text{src}(\alpha) = \text{tgt}(\alpha)$ , with quotient functor  $q$ . We say  $\mathcal{P}$  **presents** a category  $C$  when we are given an equivalence  $\langle \mathcal{P} \rangle \rightarrow C$ .

**Proposition 2.3** ([Ara+23, Ch. 2]). *Let  $\mathcal{P}$  be a 2-polygraph and  $C$  a category.*

1. *A functor  $\langle \mathcal{P} \rangle \rightarrow C$  is the same thing as a map  $V \rightarrow \text{Ob } C$  together with an arrow of  $C$  for each 1-cell, subject to the condition that the two words of every 2-cell are sent to the same arrow.*

2. Two functors out of  $\langle P \rangle$  that agree on 0-cells and 1-cells are equal.
3. A morphism of polygraphs that is bijective in every dimension is an isomorphism.

**Proposition 2.4** ([GZ67, Ch. I]). *Let  $W$  be a class of morphisms of a category  $C$ . There is a functor  $Q : C \rightarrow C[W^{-1}]$  inverting  $W$  through which every functor inverting  $W$  factors uniquely, and two functors out of  $C[W^{-1}]$  that agree after  $Q$  are equal.*

We need no calculus of fractions: the only arrows of a localization we ever write down are the composites  $Q(g^{\text{op}}) \circ Q(m^{\text{op}})^{-1}$  arising from a cospan whose first leg  $m$  lies in  $W$ , and every identity we use about them is an instance of  $Q$  being a functor.

**Newman's lemma.** Write  $\text{Hasse}(P)$  for the Hasse quiver of a poset  $P$ . A **reading** of  $P$  in a category  $C$  assigns an object to each element and an arrow to each covering relation; it extends to paths by composition.

**Theorem 2.5** (coherent Newman's lemma; [Hue80], [Ara+23, Ch. 4]). *Let  $P$  be a finite poset and  $\varphi$  a reading of  $P$  in  $C$ . Suppose that for every pair of distinct covers  $x \triangleleft y_1$  and  $x \triangleleft y_2$  and every common upper bound  $z$  of  $y_1$  and  $y_2$  there are an element  $j \leq z$  and paths  $p_i$  from  $y_i$  to  $j$  with*

$$\varphi(x \triangleleft y_1) \varphi(p_1) = \varphi(x \triangleleft y_2) \varphi(p_2).$$

*Then  $\varphi$  sends parallel paths to equal arrows, so it extends uniquely to a functor  $P \rightarrow C$ ; and a functor out of  $P$  is determined by its values on covers.*

**Lemma 2.6.** *Let  $P$  be a finite poset, graded by  $r$  with  $r(y) = r(x) + 1$  on covers, in which every element has exactly two Hasse neighbours. Then a directed path is determined by its first edge. If moreover  $P$  has a least and a greatest element, then  $P$  has exactly two maximal chains, they have a common length  $\ell$ , and  $|P| = 2\ell$ .*

*Proof.* Undirected, the Hasse graph is 2-regular, hence a disjoint union of cycles, and  $r$  increases along a directed path, so a path leaves each vertex by the edge it did not arrive on. Both edges at a least element point up and both at a greatest point down, so the cycle through the least element meets the greatest one and splits into two directed arcs, of equal length by the grading. That cycle is all of  $P$ , since every element lies on a path from the least to the greatest.  $\square$

### 3 Squares, crossings, and the braiding

Throughout,  $a$  and  $b$  are shapes on  $N$  events, read as partitions of  $[N]$  into consecutive intervals (Convention 2.1).

**Definition 3.1.** A **square** of  $a$  is a pair  $s = (p, q)$  of events with  $p < q$  and  $\text{blk}(p) = \text{blk}(q)$ ; write  $\text{Sq}(a)$  for the set of squares, so that  $|\text{Sq}(a)| = \sum_i \binom{a_i}{2}$ . A set  $R \subseteq \text{Sq}(a)$  is **biclosed** when for all events  $p < q < r$  of one block,

$$(p, q) \in R \text{ and } (q, r) \in R \implies (p, r) \in R,$$

$$(p, r) \in R \implies (p, q) \in R \text{ or } (q, r) \in R.$$

We write  $\text{Sq}(d)$  for the squares of a chain's shape, and  $f_*$  also for the induced injection  $\text{Sq}(a) \rightarrow \text{Sq}(b)$ ,  $(p, q) \mapsto (f_*p, f_*q)$ , which is well defined because  $f_*$  is increasing on blocks (Proposition 2.2).

The second clause says exactly that the complement of  $R$  is closed under the first, so the biclosed sets are closed under complementation.

**Definition 3.2.** A map of serial wedges  $f : \bigvee a \rightarrow \bigvee b$  **crosses** the squares of  $b$  whose second event it brings from a strictly earlier block of  $a$  than its first:

$$\text{Cr}(f) := \{ (p, q) \in \text{Sq}(b) : \text{blk}_a(f_*^{-1}q) < \text{blk}_a(f_*^{-1}p) \}.$$

**Lemma 3.3.** A square  $(p, q)$  of  $b$  lies in the image of  $f_*$  if and only if  $f_*^{-1}p$  and  $f_*^{-1}q$  lie in one block of  $a$ ; and for events  $u, v$  of  $b$  in distinct blocks,  $\text{blk}_a(f_*^{-1}u) < \text{blk}_a(f_*^{-1}v)$  if and only if  $\text{blk}_b(u) < \text{blk}_b(v)$ .

*Proof.* A square in the image pulls back to a square, whose events share a block; conversely if the two preimages share a block then,  $f_*$  being increasing there, they form a square mapping to  $(p, q)$ . The second claim is (1) with  $\text{blk}(f)$  monotone, read in both directions, the two blocks being distinct.  $\square$

**Lemma 3.4.** For  $f : \bigvee a \rightarrow \bigvee b$  and  $g : \bigvee b \rightarrow \bigvee c$ ,

$$\text{Cr}(gf) = g_*(\text{Cr } f) \sqcup \text{Cr}(g),$$

$a$  disjoint union; and  $\text{Cr}(\text{id}) = \emptyset$ .

*Proof.* Let  $s$  be a square of  $c$ . If  $s = g_*(t)$ , then  $s \notin \text{Cr}(g)$  because the two events of  $t$  share a block (Lemma 3.3) — which also gives disjointness — and  $(gf)_*^{-1}$  agrees with  $f_*^{-1}$  on the events of  $t$ , so  $s \in \text{Cr}(gf)$  if and only if  $t \in \text{Cr}(f)$ . If  $s$  is not in the image of  $g_*$ , then it is not in  $g_*(\text{Cr } f)$ , and its two events pull back along  $g_*$  into distinct blocks of  $b$ , so the second part of Lemma 3.3 applied to  $f$  gives  $s \in \text{Cr}(gf)$  if and only if  $s \in \text{Cr}(g)$ .  $\square$

**Definition 3.5.** A refinement  $f$  is **order preserving** when  $\text{Cr}(f) = \emptyset$ . Write  $W = W_K$  for the class of order-preserving refinements of  $\text{Ch}(K)$  and

$$\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}], \quad [c] := Q(c) \text{ for a chain } c.$$

**Proposition 3.6.** A refinement  $f$  is order preserving if and only if  $f_* = \text{id}_{[N]}$ . Consequently  $W$  contains the identities;  $gf \in W$  if and only if  $f \in W$  and  $g \in W$ ; and any two order-preserving refinements with the same source and target are equal. Moreover  $\text{Ch}(u)$  carries  $W_K$  into  $W_L$ , so it descends to a functor  $\text{Br}(u) : \text{Br}(K) \rightarrow \text{Br}(L)$ , strictly functorially in  $u$ .

*Proof.* Suppose  $f_* = \text{id}$  and let  $(p, q)$  be a square of  $b$ . Then  $\text{blk}(f)(\text{blk}_a p) = \text{blk}_b(p) = \text{blk}_b(q) = \text{blk}(f)(\text{blk}_a q)$  by (1), while  $p < q$  and  $\text{blk}_a$  monotone give  $\text{blk}_a(p) \leq \text{blk}_a(q)$ ; so  $(p, q) \notin \text{Cr}(f)$ .

Conversely suppose  $\text{Cr}(f) = \emptyset$ , and take  $p < q$  in  $[N]$ . If  $\text{blk}_a(p) = \text{blk}_a(q)$  then  $f_*p < f_*q$ , since  $f_*$  is increasing on blocks. If  $\text{blk}_a(p) < \text{blk}_a(q)$  then  $\text{blk}_b(f_*p) \leq \text{blk}_b(f_*q)$  by (1), so the only way to have  $f_*q < f_*p$  is for both to lie in one block of  $b$ ; but then  $(f_*q, f_*p)$  is a square crossed by  $f$ . So  $f_*$  is monotone, and a monotone permutation is the identity.

The three consequences are then immediate, the middle one from Lemma 3.4 with  $g_*$  injective and the last because  $f_*$  determines  $f$  (Proposition 2.2). Finally  $\text{Ch}(u)$  changes no shape and no wedge map, so it preserves the condition  $f_* = \text{id}$ ; the descent is Proposition 2.4, and it is strict because  $\text{Ch}$  is a functor.  $\square$

So an order-preserving refinement is one that only glues adjacent blocks. The elementary ones are the comparisons  $\vee[\dots, p, q, \dots] \rightarrow \vee[\dots, p + q, \dots]$ , and  $W$  is the multiplicative class they generate.

## 4 The paper polygraph

Fix  $K$ , and let  $d$  be a chain of  $K$  on  $N$  events.

### 4.1 Runs over a chain

**Definition 4.1.** A **run over  $d$**  is a refinement  $x : X \rightarrow d$  whose source  $X$  is a run. Write  $\mathcal{R}(d)$  for the set of these. Since  $X$  has shape  $[1, \dots, 1]$ , the permutation  $x_*$  enumerates the events of  $d$  in the order  $x$  fires them; we write  $p <_x q$  when  $x_*^{-1}(p) < x_*^{-1}(q)$  and call  $<_x$  the **firing order** of  $x$ .

A refinement out of a run is determined by its underlying map of wedges, the classifying map of the source being forced; so  $\mathcal{R}(d)$  is the set of runs of the wedge  $\vee a$ , and  $\text{Cr}(x)$  is the set of squares  $(p, q)$  of  $d$  with  $q <_x p$  — the squares  $x$  reverses.

**Lemma 4.2.** *A run over  $d$  fires the blocks of  $d$  in order:  $\text{blk}(p) < \text{blk}(q)$  implies  $p <_x q$ . So  $x$  is exactly a choice of linear order on each block of  $d$ , and the map  $\text{blk}$  is a monotone surjection from the firing order onto the blocks, with interval fibres.*

*Proof.* The blocks of the source are singletons, so  $\text{blk} \circ x_* = \text{blk}(x)$ , which is monotone by Proposition 2.2. A monotone surjection out of a chain has interval fibres, here the blocks of  $d$ , and the only freedom left is the order inside each one.  $\square$

**Theorem 4.3.** *The map  $x \mapsto \text{Cr}(x)$  is a bijection from  $\mathcal{R}(d)$  onto the biclosed subsets of  $\text{Sq}(d)$ . Reversing the order on every block complements  $\text{Cr}$ .*

*Proof.* Both the statement and the biclosedness condition are local to a block, by Lemma 4.2; so fix a block and let  $\prec$  be a linear order on it. Then  $\text{Cr}(\prec) = \{(p, q) : p < q, q \prec p\}$  is its inversion set, and  $\prec$  is recovered from a set  $R$  by declaring  $p \prec q$  when either  $p < q$  and  $(p, q) \notin R$ , or  $q < p$  and  $(q, p) \in R$ . So the map is injective, and its image is the set of  $R$  for which that recipe is transitive — totality being automatic. Transitivity can fail only on a triple  $p < q < r$ , and the cases to check there are exactly the two biclosedness clauses.  $\square$

For a single block of dimension  $m$  this is the classical description of the inversion sets of permutations of  $[m]$ , and inclusion of inversion sets is the right weak order [BB05, §3.1]; the order theory below is that statement, block by block. Accordingly we order  $\mathcal{R}(d)$  by

$$x \leq y \quad :\Longleftrightarrow \quad \text{Cr}(x) \subseteq \text{Cr}(y),$$

and write  $\perp_d$  and  $\top_d$  for the least and greatest elements, which exist because  $\emptyset$  and  $\text{Sq}(d)$  are biclosed.

**Proposition 4.4.** *Let  $q : o \rightarrow d$  be a refinement. Composition with  $q$  is an order embedding  $\mathcal{R}(o) \rightarrow \mathcal{R}(d)$ , and*

$$\text{Cr}(qx) = q_*(\text{Cr } x) \sqcup \text{Cr}(q).$$

*Moreover  $q\perp_o = \perp_d$  if and only if  $q \in W$ ; and for  $x \in \mathcal{R}(d)$  the conditions  $x \in W$ ,  $\text{Cr}(x) = \emptyset$  and  $x = \perp_d$  agree. In particular  $\perp_X = \text{id}_X$  for a run  $X$ , and  $\text{dom } \perp_d = \text{dom } \perp_c$  whenever  $c \rightarrow d$  lies in  $W$ .*

*Proof.* The displayed formula is Lemma 3.4. Since  $q_*$  is injective with image disjoint from  $\text{Cr}(q)$  (Lemma 3.3), it follows that  $qx \leq qy$  if and only if  $x \leq y$ ; and  $\text{Cr}(q \perp_o) = \text{Cr}(q)$ . The remaining equivalences are Theorem 4.3, and the last sentence is the formula applied to  $c \rightarrow d$ .  $\square$

## 4.2 Steps, covers, and cells

**Lemma 4.5.** *Call a square  $s$  a **step** of  $x \in \mathcal{R}(d)$  when  $\text{Cr}(x) \triangle \{s\}$  is biclosed, write  $\text{St}(x)$  for the set of steps, and write  $x^s \in \mathcal{R}(d)$  for the run with  $\text{Cr}(x^s) = \text{Cr}(x) \triangle \{s\}$ . Then  $s$  is a step of  $x$  if and only if its two events are consecutive in the firing order of  $x$ , and  $|\text{St}(x)| = \deg d$ .*

*Proof.* A block's events are consecutive in the firing order (Lemma 4.2), so “consecutive in the block” and “consecutive overall” agree. If the two events of  $s$  are consecutive, then toggling  $s$  swaps an adjacent pair of a linear order, and the result is a linear order, so it is biclosed by Theorem 4.3. If some third event of the block fires strictly between them, then toggling  $s$  swaps the pair but not the order of either event against that third one, which would have to lie both before and after the pair.

So the steps are the covers of the firing order that lie inside a block. The firing order is a chain on  $N$  elements and  $\text{blk}$  is a monotone surjection from it onto the  $r$  blocks (Lemma 4.2); such a map collapses exactly  $N - r$  covers, so there are  $N - r = \deg d$  of them.  $\square$

**Lemma 4.6.** *If  $x < z$  in  $\mathcal{R}(d)$ , then some  $s \in \text{St}(x) \setminus \text{Cr}(x)$  lies in  $\text{Cr}(z)$ . Consequently  $x < y$  if and only if  $\text{Cr}(y) = \text{Cr}(x) \cup \{s\}$  for such an  $s$ , in which case  $y = x^s$ ; so  $\mathcal{R}(d)$  is graded by  $|\text{Cr}|$ , from 0 at  $\perp_d$  to  $|\text{Sq}(d)|$  at  $\top_d$ .*

*Proof.* If  $z$  agreed with  $x$  on every pair that is covering for  $<_x$ , then  $<_x$  would be contained in  $<_z$  by transitivity, and the two linear orders would be equal, so  $z = x$ . So some covering pair  $p <_x q$  has  $q <_z p$ . Its two events share a block, since otherwise Lemma 4.2 would make every run fire  $p$  first. So the pair is a step  $s$  of  $x$ , by Lemma 4.5. If  $s = (p, q)$  then  $s \in \text{Cr}(z) \setminus \text{Cr}(x)$ , as wanted; and if  $s = (q, p)$  then  $s \in \text{Cr}(x) \subseteq \text{Cr}(z)$ , which says  $p <_z q$ , a contradiction.

For the covering criterion: given such an  $s$ , we have  $x \leq x^s \leq y$ , which forces  $y = x^s$ ; and conversely two biclosed sets differing by one element correspond to a covering pair, by Theorem 4.3.  $\square$

**Proposition 4.7.** *Let  $x \in \mathcal{R}(d)$ . Sending a factorisation  $X \rightarrow m \rightarrow d$  of  $x$  to the set of steps of  $x$  whose two events share a block of  $m$  is a bijection onto the subsets of  $\text{St}(x)$ , and  $\deg m = |S|$  for the factorisation at  $S$ . Write  $m_S$  for the middle and  $\iota_S : X \rightarrow m_S$ ,  $\pi_S : m_S \rightarrow d$  for the legs. If moreover  $S$  meets  $\text{Cr}(x)$  in nothing — we then say  $S$  **risers** from  $x$  — then  $\iota_S$  is order preserving.*

*Proof.* A middle  $m$  makes the  $N$  events into its blocks, which by Lemma 4.2 applied to  $\iota$  are intervals of the firing order of  $x$ , and  $\pi$  requires each such interval to sit inside one block of  $d$ . Conversely such an interval partition arises from the chain whose blocks are its parts, in order. The factorisation is determined by that shape: a map of wedges is determined by its action on events and injective there (Proposition 2.2), hence a monomorphism, while the first leg is forced by which events fall in which part. Now an interval partition of a chain is the same thing as the set of covers it does not cut, those covers lie inside a block, so they are steps (Lemma 4.5); and  $\deg m$  is  $N$  minus the number of parts, which is the number of uncut covers.

Suppose  $S$  rises from  $x$ . Since  $\iota_S$  is a run over  $m_S$ , its crossings are the squares of  $m_S$  that  $x$  reverses, that is the pairs inside one part that  $x$  reverses. The covers of the firing order inside a

part all lie in  $S$ , so none of them is reversed; hence the firing order agrees with the event order on consecutive pairs of the part, and so, by transitivity, on every pair of it. So  $\text{Cr}(\iota_S) = \emptyset$ .  $\square$

**Definition 4.8.** For runs  $X, Y \in \text{Run}(K)$ , an  $n$ -cell from  $X$  to  $Y$  is a chain  $e$  with  $\deg e = n$ ,  $\text{dom } \perp_e = X$  and  $\text{dom } \top_e = Y$ ; write  $\text{Cell}_n(X, Y)$  for the set of them.

So the  $n$ -cells are the chains of degree  $n$ , sorted by the two runs each one spans. A 0-cell is a run, and Proposition 4.7 says that a chain of degree  $n$  arises from any run over it and any rising set of  $n$  of that run's steps.

**Lemma 4.9.** *Let  $S$  rise from  $x \in \mathcal{R}(d)$ . Then*

$$\text{Cr}(\pi_S \top_{m_S}) = \text{Cr}(x) \cup \pi_{S*}(\text{Sq}(m_S)),$$

*and every  $z \geq x$  with  $S \subseteq \text{Cr}(z)$  satisfies  $\pi_{S*}(\text{Sq}(m_S)) \subseteq \text{Cr}(z)$ . So  $\pi_S \top_{m_S}$  is the least element of  $\mathcal{R}(d)$  above  $x$  that reverses all of  $S$ .*

*Proof.* Proposition 4.4 gives  $\text{Cr}(\pi_S \top_{m_S}) = \pi_{S*} \text{Sq}(m_S) \cup \text{Cr}(\pi_S)$ , and Lemma 3.4 applied to  $\pi_S \iota_S = x$  with  $\text{Cr}(\iota_S) = \emptyset$  gives  $\text{Cr}(\pi_S) = \text{Cr}(x)$ .

For the second claim, list a part of the partition as  $p_1 < \dots < p_k$ ; as in the previous proof this is also its firing order, so each  $(p_i, p_{i+1})$  lies in  $S$  and is reversed by  $z$ . Since  $\text{Cr}(z)$  is biclosed, the first clause applied to the triples  $(p_i, p_{i+1}, p_j)$  gives, by induction on  $j - i$ , that  $z$  reverses every pair of the part.  $\square$

**Corollary 4.10.** *Let  $x \leq y$  in  $\mathcal{R}(d)$ , with step  $s$ . Then  $\gamma(x \leq y) := m_{\{s\}}$  is a 1-cell from  $\text{dom } x$  to  $\text{dom } y$ , and  $\pi_{\{s\}} \perp_{m_{\{s\}}} = x$  and  $\pi_{\{s\}} \top_{m_{\{s\}}} = y$ . Conversely every 1-cell  $\alpha$  is  $\gamma$  of the unique cover  $\perp_\alpha \leq \top_\alpha$  of  $\mathcal{R}(\alpha)$ . Thus  $\gamma$ , together with  $x \mapsto \text{dom } x$  on vertices, is a map of quivers from  $\text{Hasse}(\mathcal{R}(d))$  to the quiver of 1-cells, and  $\gamma(qe) = \gamma(e)$  for every refinement  $q : o \rightarrow d$  and every cover  $e$  of  $\mathcal{R}(o)$ .*

*Proof.* The chain  $m_{\{s\}}$  has degree 1 and  $\text{dom } \perp_{m_{\{s\}}} = \text{dom } x$  by Proposition 4.7; by Lemma 4.9 its top pushes forward to the least element above  $x$  reversing  $s$ , namely  $x^s = y$ . Conversely a chain  $\alpha$  of degree 1 has one square, so  $\mathcal{R}(\alpha)$  is the two-element poset  $\perp_\alpha \leq \top_\alpha$  and the factorisation through its one step is  $\alpha$  itself. For the last claim, the step of  $qe$  is  $q_*$  of the step of  $e$  (Proposition 4.4), and  $q_*$  carries the firing order of  $x$  to that of  $qx$ , so the two factorisations cut the same partition and have the same middle.  $\square$

### 4.3 The attachment of a 2-cell, and the polygraph

**Lemma 4.11.** *Order  $\text{Sq}(d)$  lexicographically, first by the smaller event and then by the larger. Every lower set of that order is biclosed, and so is every upper set. Consequently, if  $\prec$  is either that order or its reverse, then the runs  $x_t$  with  $\text{Cr}(x_t)$  the first  $t$  squares form a maximal chain  $\perp_d = x_0 \leq \dots \leq x_{|\text{Sq}(d)|} = \top_d$ , in which the  $(t+1)$ -st square is a step of  $x_t$ .*

*Proof.* On a triple  $p < q < r$  of one block the three squares come in the order  $(p, q) \prec (p, r) \prec (q, r)$ . If  $R$  is a lower set then  $(q, r) \in R$  forces  $(p, r) \in R$ , and  $(p, r) \in R$  forces  $(p, q) \in R$ , which are the two clauses; if  $R$  is an upper set then  $(p, q) \in R$  forces  $(p, r) \in R$  and  $(p, r) \in R$  forces  $(q, r) \in R$ . The chain then exists by Theorem 4.3, and consecutive terms differ by one element, so Lemma 4.6 applies.  $\square$

**Definition 4.12.** For a chain  $e$  with  $\deg e = 2$ , let  $\text{src}(e)$  and  $\text{tgt}(e)$  be the words of 1-cells that the two chains of Lemma 4.11 read off under  $\gamma$ , for the lexicographic order and for its reverse respectively.

**Proposition 4.13.** *Let  $e$  be a chain with  $\deg e = 2$ . Then  $\mathcal{R}(e)$  is a polygon with  $2|\text{Sq}(e)|$  elements, whose only two maximal chains from  $\perp_e$  to  $\top_e$  are the two of Lemma 4.11; so  $\text{src}(e)$  and  $\text{tgt}(e)$  have length  $|\text{Sq}(e)|$  and differ in their first letter. Moreover  $e$  has either one block of dimension 3, in which case  $|\text{Sq}(e)| = 3$  and  $\mathcal{R}(e)$  is a hexagon, or two blocks of dimension 2, in which case  $|\text{Sq}(e)| = 2$  and  $\mathcal{R}(e)$  is a square.*

*Proof.* Every  $x \in \mathcal{R}(e)$  has exactly two steps by Lemma 4.5, and the two flips are its two Hasse neighbours, upward or downward according to whether the step is already reversed (Lemma 4.6). The two chains of Lemma 4.11 leave  $\perp_e$  by different covers, the least and the greatest square being distinct, so Lemma 2.6 applies with the grading  $|\text{Cr}|$ . Finally  $\deg e = 2$  forces the block dimensions above 1 to be either  $\{3\}$  or  $\{2, 2\}$ , and Definition 3.1 counts  $\binom{3}{2} = 3$  and  $2\binom{2}{2} = 2$ .  $\square$

**Definition 4.14.** The **paper polygraph**  $\mathcal{P}(K)$  has 0-cells the runs of  $K$ , 1-cells  $\mathcal{P}(K)_1(X, Y) := \text{Cell}_1(X, Y)$ , 2-cells  $\mathcal{P}(K)_2(X, Y) := \text{Cell}_2(X, Y)$ , and the two words of Definition 4.12.

Its data is nothing but the chains of  $K$  of degree 0, 1 and 2. The only computation is the attachment, and by Proposition 4.13 that computation has exactly two answers.

**Proposition 4.15.** *Read in  $\langle \mathcal{P}(K) \rangle$ , the quiver map  $\gamma$  satisfies the hypothesis of Theorem 2.5 on  $\mathcal{R}(d)$ . So it extends uniquely to a functor  $\Gamma_d : \mathcal{R}(d) \rightarrow \langle \mathcal{P}(K) \rangle$ , and  $\Gamma_d(qx) = \Gamma_o(x)$  for every refinement  $q : o \rightarrow d$  and every  $x \in \mathcal{R}(o)$ , covers included.*

*Proof.* Let  $x \leq y_1$  and  $x \leq y_2$  be distinct covers, with steps  $s_1 \neq s_2$ , and let  $z$  be a common upper bound. Then  $S = \{s_1, s_2\}$  rises from  $x$ , so by Proposition 4.7 the middle  $\alpha := m_S$  is a chain of degree 2, that is a 2-cell, and  $j := \pi_S \top_\alpha$  satisfies  $j \leq z$  by Lemma 4.9. By Proposition 4.13 the two maximal chains of  $\mathcal{R}(\alpha)$  are  $\text{src}(\alpha)$  and  $\text{tgt}(\alpha)$ , and they leave  $\perp_\alpha$  by its two covers. Composition with  $\pi_S$  carries  $\perp_\alpha$  to  $x$  and  $\top_\alpha$  to  $j$ , carries covers to covers (Proposition 4.4, Lemma 4.6) and keeps the 1-cell each cover reads as (Corollary 4.10). So the two chains become two paths from  $x$  to  $j$ , leaving by  $y_1$  and by  $y_2$  and reading as  $\text{src}(\alpha)$  and  $\text{tgt}(\alpha)$ , which are equal in  $\langle \mathcal{P}(K) \rangle$ .

For the last claim, composition with  $q$  is monotone, so  $x \mapsto \Gamma_d(qx)$  and  $\Gamma_o$  are functors out of  $\mathcal{R}(o)$ ; they agree on covers by Corollary 4.10, hence agree, by the last clause of Theorem 2.5.  $\square$

## 5 Theorem A: the polygraph presents $\text{Br}(K)$

Theorem A compares two universal properties. A functor out of  $\langle \mathcal{P}(K) \rangle$  is a reading of the degree-one chains that identifies the two words of every degree-two chain (Proposition 2.3); a functor out of  $\text{Br}(K)$  is a functor on  $\text{Ch}(K)^{\text{op}}$  inverting the order-preserving refinements (Proposition 2.4). We build each from the other.

A cell  $\alpha \in \text{Cell}_n(X, Y)$  comes with two refinements into it, namely the least and greatest runs over it,  $\perp_\alpha : X \rightarrow \alpha$  and  $\top_\alpha : Y \rightarrow \alpha$ , of which the first is order preserving. Since  $Q$  inverts  $\perp_\alpha^{\text{op}}$ , we may set

$$E(X) := [X], \quad E(\alpha) := Q(\top_\alpha^{\text{op}}) \circ Q(\perp_\alpha^{\text{op}})^{-1} : [X] \rightarrow [Y].$$

**Lemma 5.1.** *Let  $d$  be a chain and  $z \in \mathcal{R}(d)$ . Every path from  $\perp_d$  to  $z$  in  $\text{Hasse}(\mathcal{R}(d))$  reads, under  $E \circ \gamma$ , as the arrow  $Q(z^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1}$ .*

*Proof.* Induct on the length of the path; the empty path reads as the identity. For a cover  $x \leq y$  with step  $s$ , put  $\beta := \gamma(x \leq y)$  and  $q := \pi_{\{s\}}$ , so that  $x = q \perp_\beta$  and  $y = q \top_\beta$  by Corollary 4.10. Then  $x^{\text{op}}$  is  $\perp_\beta^{\text{op}} \circ q^{\text{op}}$ , so the middle two factors of

$$E(\beta) \circ Q(x^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1} = Q(\top_\beta^{\text{op}}) \circ Q(\perp_\beta^{\text{op}})^{-1} \circ Q(\perp_\beta^{\text{op}}) \circ Q(q^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1}$$

cancel, leaving  $Q(\top_\beta^{\text{op}})Q(q^{\text{op}})Q(\perp_d^{\text{op}})^{-1} = Q(y^{\text{op}})Q(\perp_d^{\text{op}})^{-1}$ .  $\square$

**Corollary 5.2.** *The two words of a 2-cell have the same image under  $E \circ \gamma$ , so  $E$  extends to a functor  $E : \langle \mathcal{P}(K) \rangle \rightarrow \text{Br}(K)$ .*

*Proof.* Both words are paths from  $\perp_e$  to  $\top_e$  (Proposition 4.13), so both read as  $Q(\top_e^{\text{op}})Q(\perp_e^{\text{op}})^{-1}$  by Lemma 5.1. Now apply Proposition 2.3(i).  $\square$

In the other direction, define  $\Theta : \text{Ch}(K)^{\text{op}} \rightarrow \langle \mathcal{P}(K) \rangle$  on objects by  $\Theta(d) := \text{dom } \perp_d$  and on a refinement  $u : c \rightarrow d$  by

$$\Theta(u^{\text{op}}) := \Gamma_d(\perp_d \leq u \perp_c) : \text{dom } \perp_d \rightarrow \text{dom } \perp_c,$$

which makes sense because  $u \perp_c \in \mathcal{R}(d)$  has source  $\text{dom } \perp_c$ .

**Lemma 5.3.**  *$\Theta$  is a functor, and  $\Theta(u^{\text{op}})$  is an identity whenever  $u$  is order preserving. So  $\Theta$  descends uniquely to a functor  $\Phi : \text{Br}(K) \rightarrow \langle \mathcal{P}(K) \rangle$  with  $\Phi Q = \Theta$ .*

*Proof.* Let  $v : b \rightarrow c$  and  $u : c \rightarrow d$ . Composition with  $u$  is monotone (Proposition 4.4), so  $\perp_d \leq u \perp_c \leq uv \perp_b$ , and since  $\Gamma_d$  is a functor,  $\Theta((uv)^{\text{op}}) = \Gamma_d(u \perp_c \leq uv \perp_b) \circ \Theta(u^{\text{op}})$ ; the left factor is  $\Gamma_c(\perp_c \leq v \perp_b) = \Theta(v^{\text{op}})$  by the last clause of Proposition 4.15. If  $u$  is order preserving then  $u \perp_c = \perp_d$  by Proposition 4.4, so the comparison, and hence  $\Theta(u^{\text{op}})$ , is an identity. The descent is then Proposition 2.4.  $\square$

**Theorem 5.4.** *For every bipointed precubical set  $K$ , the functor  $E$  is an equivalence  $\langle \mathcal{P}(K) \rangle \rightarrow \text{Br}(K)$ , with inverse  $\Phi$ .*

*Proof.* We show  $\Phi E$  is the identity. On a 0-cell,  $\Phi(E(X)) = \Theta(X) = \text{dom } \perp_X = X$  by Proposition 4.4. On a 1-cell  $\alpha$ , applying  $\Phi$  to the defining composite gives  $\Phi(E(\alpha)) = \Theta(\top_\alpha^{\text{op}}) \circ \Theta(\perp_\alpha^{\text{op}})^{-1}$ , and the second factor is an identity because  $\perp_\alpha$  is order preserving (Lemma 5.3). Now  $\top_\alpha \perp_Y = \top_\alpha$ , and  $\mathcal{R}(\alpha)$  is the two-element poset  $\perp_\alpha \leq \top_\alpha$ , so

$$\Theta(\top_\alpha^{\text{op}}) = \Gamma_\alpha(\perp_\alpha \leq \top_\alpha) = q(\gamma(\perp_\alpha \leq \top_\alpha)) = q(\alpha)$$

by Corollary 4.10. So  $\Phi E$  agrees with the identity on 0- and 1-cells, hence is the identity by Proposition 2.3(ii).

For the other composite, Lemma 5.1 gives  $E(\Theta(u^{\text{op}})) = Q((u \perp_c)^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1} = Q(\perp_c^{\text{op}}) \circ Q(u^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1}$ , which is  $Q(u^{\text{op}})$  conjugated by the isomorphisms  $[c] \cong [\text{dom } \perp_c]$  that  $Q$  makes of the order-preserving refinements  $\perp_c$ . So  $E\Theta \cong Q$ , and  $E\Phi \cong \text{id}$  by uniqueness of descent (Proposition 2.4).  $\square$

The comparison is an equivalence and not an isomorphism because  $\text{Br}(K)$  is not skeletal: a chain and the run below it become isomorphic there without being equal. The runs are a skeleton of  $\text{Br}(K)$ .

## 6 Theorem B: naturality in $K$

**Theorem 6.1.** *Sending  $K$  to  $\mathcal{P}(K)$ , and  $f : K \rightarrow L$  to  $\text{Ch}(f)$  on cells in every dimension, is a functor  $\square_* \rightarrow \mathbf{Poly}$ ; and for every  $f$ ,*

$$E_L \circ \langle \mathcal{P}(f) \rangle = \text{Br}(f) \circ E_K : \langle \mathcal{P}(K) \rangle \longrightarrow \text{Br}(L).$$

*Proof.* Write  $f_\bullet := \text{Ch}(f)$ . A run over a chain  $e$  is determined by its map of wedges (Definition 4.1), and  $f_\bullet e$  has the same shape and the same wedge maps into it, so  $\mathcal{R}(e) = \mathcal{R}(f_\bullet e)$  as ordered sets. Everything in Definition 3.2, Lemma 4.5 and Corollary 4.10 is defined from the wedge map alone, so  $f_\bullet$  preserves  $\perp$ ,  $\top$ , cells, steps, covers and  $\gamma$ ; and it carries the lexicographic order on  $\text{Sq}(e)$  to the lexicographic order on  $\text{Sq}(f_\bullet e) = \text{Sq}(e)$ , so it preserves  $\text{src}$  and  $\text{tgt}$  letter by letter. Hence  $\mathcal{P}(f)$  is a morphism of polygraphs, functorial in  $f$  because  $\text{Ch}$  is a functor.

Both composites send a run  $X$  to  $[f_\bullet X]$ . On a 1-cell  $\alpha$ , the functor  $\text{Br}(f)$  is the descent of  $Q_L \circ f_\bullet^{\text{op}}$ , so it sends  $E_K(\alpha)$  to  $Q((f_\bullet \top_\alpha)^{\text{op}}) \circ Q((f_\bullet \perp_\alpha)^{\text{op}})^{-1}$ . Here  $f_\bullet \perp_\alpha$  is an order-preserving refinement into  $f_\bullet \alpha$  out of a run, so it is  $\perp_{f_\bullet \alpha}$  by Proposition 4.4, and  $f_\bullet \top_\alpha = \top_{f_\bullet \alpha}$  likewise. So both composites send  $\alpha$  to  $E_L(f_\bullet \alpha)$ , and Proposition 2.3(ii) finishes the proof.  $\square$

The square commutes strictly, which is more than a presentation is entitled to. The reason is Proposition 3.6: an order-preserving refinement is determined by its source and target, so transporting the two legs of a cell involves no choice.

## 7 Theorem C: at the point the polygraph is Artin's

Since  $Z$  is terminal, every serial wedge has exactly one bipointed map to it and every map of wedges commutes over it, so  $\text{Ch}(Z)$  is the category of shapes and maps of wedges. A run of  $Z$  is therefore determined by its event count; write  $Z_N$  for the run on  $N$  events. Both  $\perp_e$  and  $\top_e$  have source a run on  $|e|$  events, so every cell of  $Z$  is a loop at  $Z_{|e|}$ .

**Definition 7.1.** For a shape  $a$  on  $N$  events put  $\text{cut}(a) := \{k \in [N-1] : \text{blk}_a(k) = \text{blk}_a(k+1)\}$ , and write  $\text{cut}(d)$  for the cuts of a chain's shape. For  $i \neq k$  in  $[N-1]$  let  $\text{cox}(i, k) := 3$  if  $|i - k| = 1$  and 2 otherwise, the Coxeter exponent of  $S_N$ .

**Lemma 7.2.** *The map  $a \mapsto \text{cut}(a)$  is a bijection from the shapes on  $N$  events onto the subsets of  $[N-1]$ , with  $|\text{cut}(a)| = \deg a$ . Hence  $\text{Cell}_n(Z_N, Z_N)$  is the set of  $n$ -element subsets of  $[N-1]$ : the 1-cells at  $Z_N$  are indexed by  $[N-1]$ , and we write  $\sigma_k$  for the one with  $\text{cut} = \{k\}$ ; the 2-cells are indexed by the pairs  $i < k$ . Moreover a degree-two chain  $e$  with  $\text{cut}(e) = \{i < k\}$  has  $|\text{Sq}(e)| = \text{cox}(i, k)$ .*

*Proof.* A shape on  $N$  events is a composition of  $N$  and  $\text{cut}(a)$  is the complement of its set of block boundaries in  $[N-1]$ , so the first claim is the classical correspondence [Sta12, §1.2], and  $|\text{cut}(a)| = N - r = \deg a$ . For the last claim,  $e$  has a block of dimension 3 exactly when three consecutive events share a block, that is exactly when  $k = i + 1$ ; then  $|\text{Sq}(e)| = 3$ , and otherwise  $|\text{Sq}(e)| = 2$ , by Proposition 4.13.  $\square$

**Lemma 7.3.** *Let  $d$  be a chain on  $N$  events and  $x \in \mathcal{R}(d)$ . For a step  $s$  of  $x$ , let  $\text{pos}_x(s)$  be the earlier of the two times at which  $x$  fires the events of  $s$ , so that  $x$  passes  $s$  between times  $\text{pos}_x(s)$  and  $\text{pos}_x(s) + 1$ . Then*

1.  $\text{pos}_x$  is a bijection from  $\text{St}(x)$  onto  $\text{cut}(d)$ ;
2.  $\text{pos}_{x^s}(s) = \text{pos}_x(s)$ ;
3.  $\text{pos}_{\perp_d}(s) = p$  for  $s = (p, q)$ .

*If moreover  $d$  is a chain of  $Z$ , then a cover  $x \leq y$  with step  $s$  has  $\gamma(x \leq y) = \sigma_{\text{pos}_x(s)}$ .*

*Proof.* (i) The fibres of  $\text{blk}$  on the firing order are consecutive intervals of lengths  $a_1, \dots, a_r$  (Lemma 4.2), which do not depend on  $x$ ; so the set of times  $t$  with  $t$  and  $t + 1$  in one block is  $\text{cut}(d)$  whatever  $x$  is. A step is a cover of the firing order inside a block (Lemma 4.5), so  $\text{pos}_x$  lands in  $\text{cut}(d)$ , and it is injective because distinct covers of a chain have distinct lower ends; both sides have  $\deg d$  elements by Lemma 4.5 and Lemma 7.2. (ii) Flipping  $s$  exchanges the two times and changes no other. (iii)  $\perp_d$  fires the events in their own order. For the last claim,  $\gamma(x \leq y) = m_{\{s\}}$ , whose blocks are the parts of the firing order cut at every cover but  $s$  (Proposition 4.7); so  $m_{\{s\}}$  has one block of dimension 2, at times  $\text{pos}_x(s)$  and  $\text{pos}_x(s) + 1$ , and its other blocks are edges, so its cut set is  $\{\text{pos}_x(s)\}$ .  $\square$

**Proposition 7.4.** *Let  $e$  be a chain of  $Z$  with  $\deg e = 2$  and  $\text{cut}(e) = \{i < k\}$ , and let  $\perp_e = x_0 \leq \dots \leq x_{|\text{Sq}(e)|} = \top_e$  be the chain of Lemma 4.11 for the lexicographic order, with  $s_t$  the  $(t + 1)$ -st square. Then  $\text{pos}_{x_t}(s_t)$  is  $i$  for  $t$  even and  $k$  for  $t$  odd. Hence*

$$\text{src}(e) = \sigma_i \sigma_k \sigma_i \cdots, \quad \text{tgt}(e) = \sigma_k \sigma_i \sigma_k \cdots,$$

*both of length  $\text{cox}(i, k)$ .*

*Proof.* By Lemma 7.3(i) every  $x_t$  has exactly two steps, passed at  $i$  and at  $k$ .

For  $t = 0$ : by Lemma 7.3(iii) the run  $\perp_e$  passes  $(p, q)$  at  $p$ , and the lexicographic order sorts  $\text{Sq}(e)$  by the smaller event first. If  $e$  is one block of dimension 3 on events  $i, i + 1, i + 2$ , its squares in order are  $(i, i + 1)$ ,  $(i, i + 2)$ ,  $(i + 1, i + 2)$ , with smaller events  $i, i, i + 1 = k$ ; if  $e$  is two blocks of dimension 2 on  $\{i, i + 1\}$  and  $\{k, k + 1\}$ , they are  $(i, i + 1)$  and  $(k, k + 1)$ , with smaller events  $i$  and  $k$ . Either way the least square is passed at  $i$  and the greatest at  $k$ .

For the induction step, both  $s_t$  and  $s_{t+1}$  are steps of  $x_{t+1}$ : the first because a flip of a step is a step, the second by Lemma 4.11. By Lemma 7.3(ii) we have  $\text{pos}_{x_{t+1}}(s_t) = \text{pos}_{x_t}(s_t)$ , so  $\text{pos}_{x_{t+1}}(s_{t+1})$  is whichever of  $i$  and  $k$  that is not. The words now follow from the last clause of Lemma 7.3, and their length is  $|\text{Sq}(e)| = \text{cox}(i, k)$  by Lemma 7.2. The reverse order has the lexicographic order's greatest square first, passed at  $k$ , and the same induction applies.  $\square$

**Definition 7.5. Artin's polygraph  $\mathcal{P}_{\text{Artin}}$**  has one 0-cell for each  $N \in \mathbb{N}$ ; at  $N$ , the loops  $\sigma_1, \dots, \sigma_{N-1}$ ; and, for each pair  $i < k$  in  $[N - 1]$ , one 2-cell whose two words are  $\sigma_i \sigma_k \sigma_i \cdots$  and  $\sigma_k \sigma_i \sigma_k \cdots$ , both of length  $\text{cox}(i, k)$ .

The relations are  $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$  when  $k = i + 1$  and  $\sigma_i \sigma_k = \sigma_k \sigma_i$  otherwise, which is Artin's presentation of the positive braid monoid  $B_N^+$  [Art47]. Path composition reverses concatenation of words, so  $\langle \mathcal{P}_{\text{Artin}} \rangle$  has endomorphism monoid  $(B_N^+)^{\text{op}}$  at  $N$  and no arrows between distinct objects; that is,  $\langle \mathcal{P}_{\text{Artin}} \rangle = \mathbf{Braid}^{+, \text{op}}$ .

**Theorem 7.6.**  $\mathcal{P}(Z) \cong \mathcal{P}_{\text{Artin}}$ .

*Proof.* The runs of  $Z$  are the  $Z_N$ , and every cell of  $Z$  is a loop, so both polygraphs are disjoint unions over  $N$  and it is enough to compare at each  $N$ . There the 1-cells are indexed by  $[N - 1]$  and the 2-cells by the pairs  $i < k$ , by Lemma 7.2, matching Artin’s generators and relations; and by Proposition 7.4 the two words of the 2-cell at  $\{i < k\}$  are Artin’s two words for that pair. A morphism of polygraphs bijective in every dimension is an isomorphism (Proposition 2.3(iii)).  $\square$

**Corollary 7.7.**  $\text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$ .

*Proof.* Theorem 5.4 at  $K = Z$ , together with Theorem 7.6.  $\square$

Combining this with Theorem 6.1, the terminal map gives  $\text{Br}(!_K) : \text{Br}(K) \rightarrow \text{Br}(Z)$ , a braid-valued invariant of the arrows of  $\text{Br}(K)$  that is computed cell by cell: a chain of degree 1 goes to the generator  $\sigma_k$ , where  $k$  is the position at which its run passes its one step, and a chain of degree 2 goes to the Artin relation of its two positions. This is the promised sharpening of [PZ21]: the braid is read off the chains of  $K$  directly, with no nerve and no realization. Matsumoto’s theorem, which identifies  $B_N^+$  with the Garside germ, plays no role, since  $B_N^+$  is taken here to be Artin’s presentation.

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